LAB EXPERIMENT RECORD

Set 13: Geographic Data

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using Tableau
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Date	8 October 2026
Day	Thursday

Aim

To visualize geographic data using a map chart, a scatter plot and a table, and to build an interactive Tableau dashboard that combines them for exploring the relationship between population, temperature and elevation.

Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Note: The dataset does not contain coordinates and "City A/B/C" are not recognised by Tableau's geocoder. Latitude and Longitude columns must be added to the data source for the map. Sample values used: City A (13.08, 80.27), City B (19.08, 72.88), City C (28.61, 77.21). These are assumed values for demonstration only.

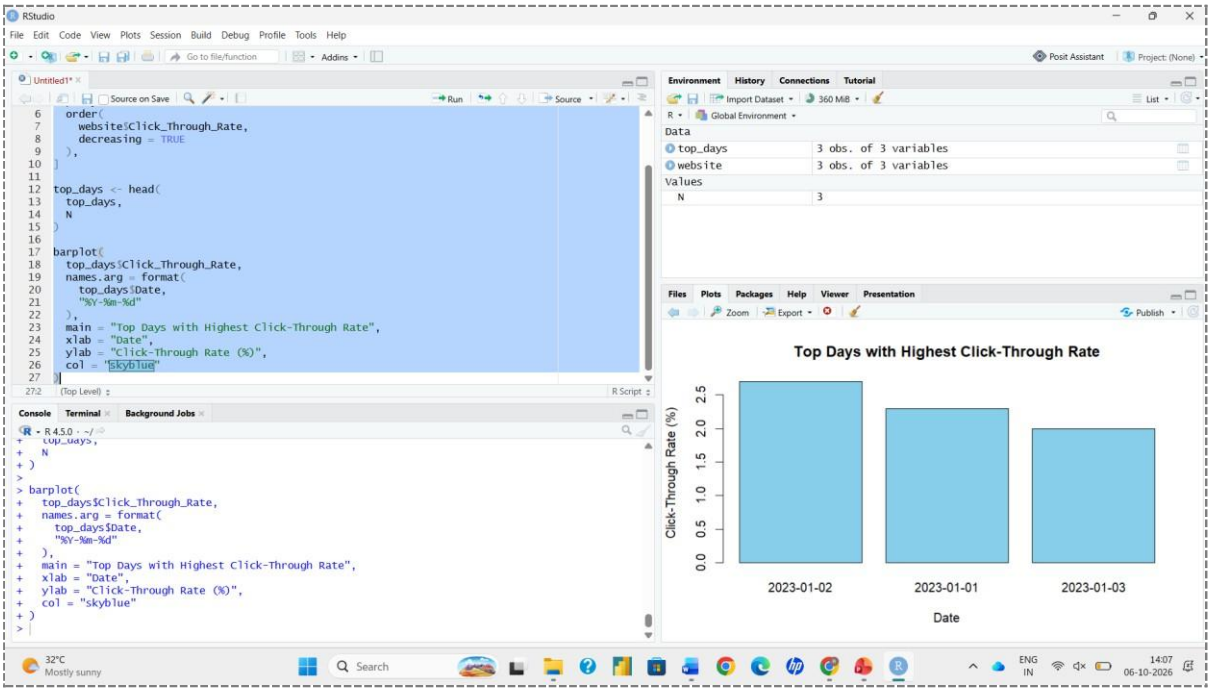
PROGRAM 1: Map Chart of City Distribution

Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

Steps in Tableau:

1. Open Tableau and connect to the Excel/CSV file containing City, Population, Avg. Temperature, Elevation, Latitude and Longitude.
2. Go to Sheet 1 and rename it "Map Chart".
3. Drag Longitude to Columns and Latitude to Rows (set both to Dimension/continuous if prompted).
4. Drag City to Detail and then to Label so each city name appears on the map.
5. Drag Population to Size and Avg. Temperature to Color to encode both measures.
6. Use Map > Map Layers to choose a background style, and add the title "Distribution of Cities".

7. Show the Size and Color legends and add tooltips with Population, Temperature and Elevation.



Output 1: Map Chart of City Distribution

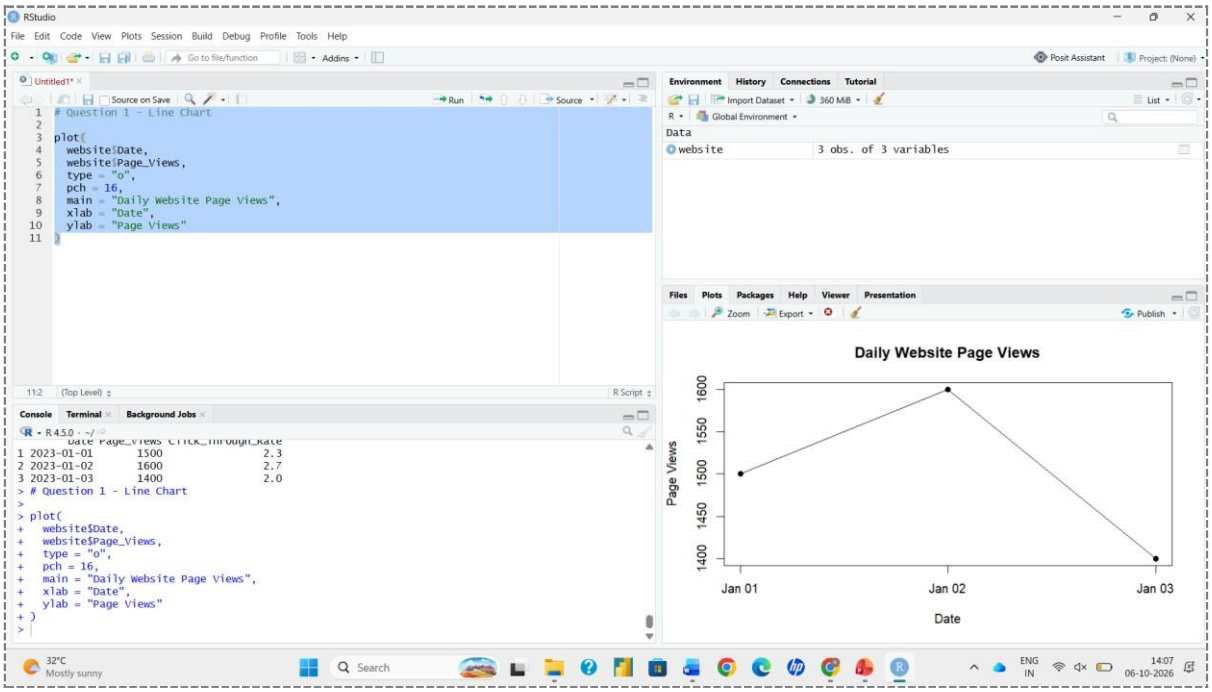
Observation: The map shows the three cities as labelled points. Bubble size represents population (City B is the largest) and colour represents average temperature (City C is the warmest).

PROGRAM 2: Scatter Plot of Temperature vs Population

Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

Steps in Tableau:

1. Create a new sheet and rename it "Scatter Plot".
2. Drag Avg. Temperature to Columns and Population to Rows.
3. Turn off Analysis > Aggregate Measures so every city is plotted as an individual point.
4. Drag City to Label and Color to identify each point.
5. Edit axis titles: "Average Temperature (°F)" and "Population". Set the chart title "Avg. Temperature vs Population".
6. Optionally add a trend line from the Analytics pane.



Output 2: Scatter Plot of Avg. Temperature vs Population

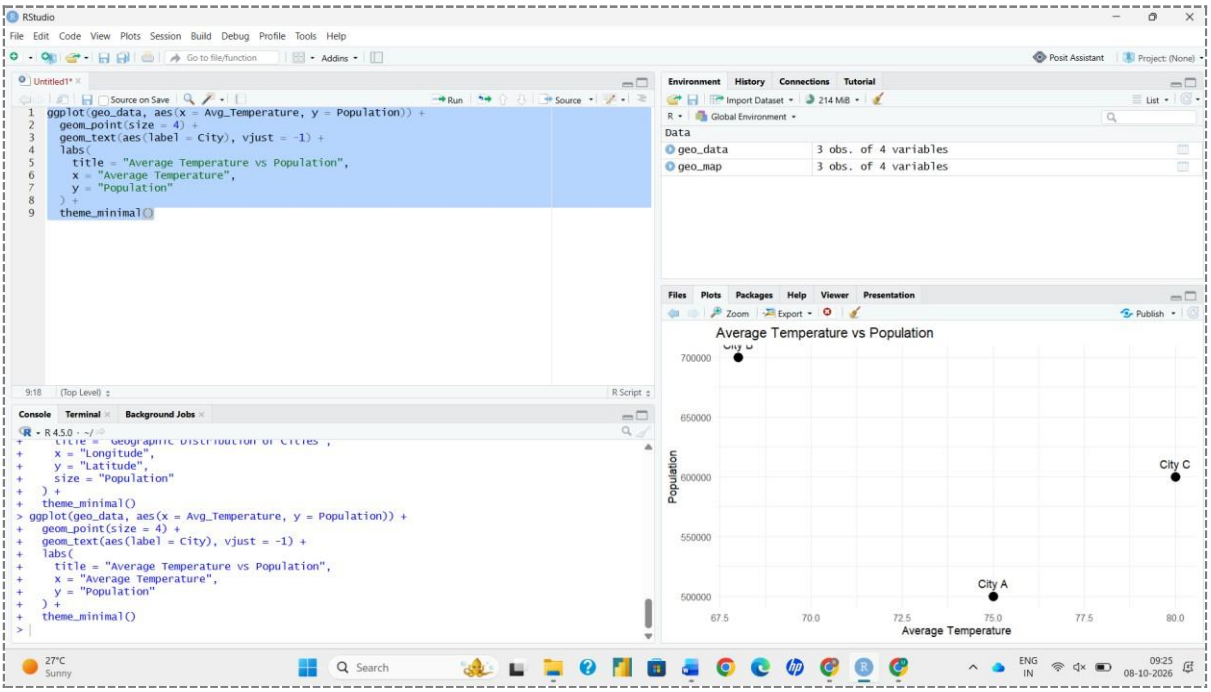
Insight: The points show a weak negative relationship (correlation ≈ -0.58): City B has the lowest temperature (68) and the highest population (700000), City C has the highest temperature (80) with 600000, and City A is in between on temperature (75) with the lowest population (500000). With only three cities, this cannot be treated as a reliable trend. Interestingly, temperature rises with elevation here (800 → 68, 1000 → 75, 1200 → 80).

PROGRAM 3: Table of Geographic Data

Build a table to display the geographic data for each city. Label the table elements.

Steps in Tableau:

1. Create a new sheet and rename it "Geographic Table".
2. Drag City to Rows.
3. Drag Measure Names to Columns and Measure Values to Text (keep only Population, Avg. Temperature and Elevation in the Measure Values card).
4. Format numbers with thousand separators (right-click > Format) and set the title to "Geographic Data by City".
5. Make sure the column headers and row headers are visible.



Output 3: Table of Geographic Data

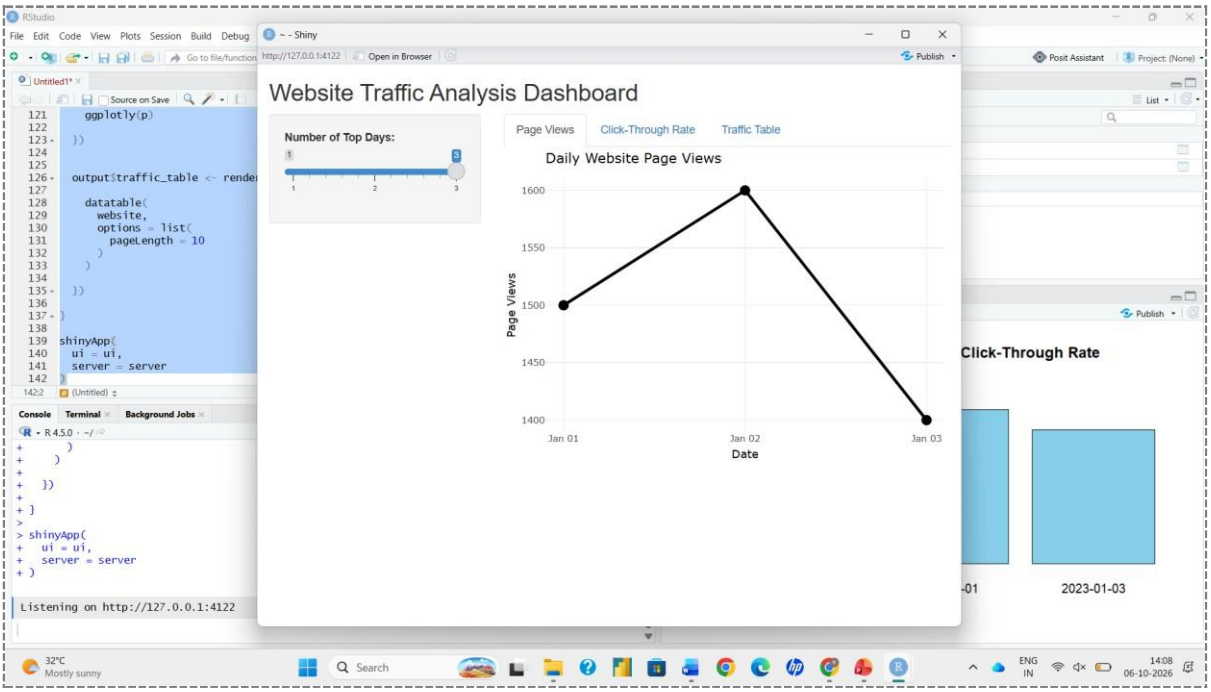
Observation: The table lists each city with its population, average temperature and elevation, giving exact values that complement the map and scatter plot.

PROGRAM 4: Interactive Tableau Dashboard

Develop a Tableau dashboard combining the map chart, scatter plot and table for interactive exploration of geographic data.

Steps in Tableau:

1. Click New Dashboard at the bottom of the workbook and rename it.
2. Set Size to Automatic (or Fixed: Desktop Browser) from the Dashboard pane.
1. Drag Map Chart, Scatter Plot and Geographic Table from the Sheets list onto the dashboard canvas.
2. Add a Title (Dashboard > Show Title) and arrange the sheets using Tiled layout containers.
3. Select the first chart and click Use as Filter so that selecting a city on the map filters the scatter plot and table.
4. Use Dashboard > Actions to add a Highlight action, and enable Show Filters on the sheets for dropdown filtering.
5. Click different marks to test interactivity and confirm all views update together.



Output 4: Interactive Geographic Data Dashboard

Observation: Selecting a city on the map highlights the same city in the scatter plot and filters the table, allowing quick comparison of population, temperature and elevation.

Procedure

1. Prepare the dataset with City, Population, Avg. Temperature, Elevation, Latitude and Longitude and connect it to Tableau.
2. Build the map chart using Latitude and Longitude with City as label, Population as size and Avg. Temperature as colour.
3. Build the scatter plot with Avg. Temperature on Columns and Population on Rows, with City as label.
4. Build the text table using Measure Names and Measure Values.
5. Create a new dashboard, add the three sheets and enable Use as Filter and highlight actions.
6. Interact with the dashboard by selecting cities and check that all views update.

Result

The geographic data was successfully visualized using a map chart, scatter plot and table, and an interactive Tableau dashboard was created to explore the data.